Mingxi Yu

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教育背景

南方科技大学（SUSTech） 2024 级本科生
计算机科学与工程系 | 计算机科学与技术 预计 2028 年 6 月毕业
学分绩点：3.76 / 4.00 **加权平均分：**89.16

项目经历

基于扩散与流模型的眼动轨迹生成 2026.8 - 至今
Computational Linguistics and Consciousness Sciences Lab (CLCS), SUSTech

- 面向基于 DDPM 的 ScanDL 2.0 的高采样开销及长文本建模难，探索生成加速、长序列眼动建模与跨数据集适配。
- 计划将原有 DDPM 采样流程改造为 DDIM，并基于 OneStop 眼动数据集微调预训练模型、扩展长文本建模能力，系统评估采样步数、生成质量与推理效率之间的权衡。
- 后续将进一步探索文本条件 Flow Matching 眼动生成模型，学习从简单先验分布到眼动轨迹分布的条件速度场，并通过 ODE 积分研究少步采样下的高效轨迹生成。

道路驾驶场景下的多模态视频文本检索与视频切片 2026.6 - 2026.8
Research Institute of Trustworthy Autonomous Systems (RITAS), SUSTech

- 主要负责道路驾驶场景多模态检索与视频切片模型设计、训练与评测，基于 Qwen3-VL-Embedding 构建视频–文本双向检索系统，使用 InfoNCE 与 LoRA 进行微调，并实现 Qwen3-VL-Reranker Top-K 两阶段重排，初步两阶段实验在 168 个验证视频上取得 **89.9%** 的 Text-to-Video Recall@1 和 **85.7%** 的 Video-to-Text Recall@1。
- 通过轨迹增强多模态检索能力，设计基于 Transformer 的车辆轨迹编码器，将轨迹信息注入 Qwen 视觉 token hidden states，实现视觉、文本与车辆轨迹的三模态联合表征学习，并与 Qwen LoRA 进行多阶段的联合训练，在单阶段 embedding 检索设置下，将 Text-to-Video 和 Video-to-Text Recall@1 分别从 80.4%/80.4% 提升至 **82.1%/81.5%（+1.7/+1.1）**。
- 在视频文本检索之外，将模型扩展至 Text-to-Clip 视频切片任务，基于行为标签与时间邻域构造强正例、邻近正例及同类难负例进行 LoRA 微调，并通过相似度筛选与 Gap Filling 将相关帧聚合为连续片段；在固定评测子集上，mean temporal IoU 从 0.333 提升至 **0.406（+21.9%）**，Frame-set Precision 从 42.7% 提升至 **48.3%**。

频域引导的轻量级小目标检测 *共同第一作者 • AAAI 在投 • arXiv:2606.23825* 2025.12 - 2026.3
Research Institute of Trustworthy Autonomous Systems (RITAS), SUSTech

- 针对小目标在下采样、特征融合与定位回归过程中高频边界信息易衰减的问题，项目构建频域引导的特征表示框架，以 WDG、LGE 与 FDHead 分别在 backbone、neck 与 detection head 中进行频率感知建模。
- 主要负责将源自 CNN 检测器的频域增强模块适配至 RT-DETRv2 的 CNN–Transformer 混合检测流程：在 YOLO11 主干 P2/P3 阶段的 C3k2_WDC 块中实现 WDG，在 HybridEncoder 的 P4/S4（stride 16）输入投影后接入轻量化 LGE，并设计面向小目标的面积偏置 Top-K query selection，完成多尺度特征接口适配与跨数据集训练验证。
- 在 RT-DETR-R18/R50 上完成 VisDrone2019、UAVDT、TinyPerson 与 DOTAv1 跨数据集验证；其中 RT-DETR-R18+ 在 UAVDT val 上取得 **+7.5 $\mathbf{mAP}_{50}$**，同时参数量与 GFLOPs 分别降低 **31.3%** 与 **25.8%**，验证频域机制的跨架构泛化与轻量化能力。

基于知识蒸馏的视频文本检索 2025.9 - 2025.11
Research Institute of Trustworthy Autonomous Systems (RITAS), SUSTech

- 面向视频–文本检索模型推理开销较高的问题，基于 TeachCLIP 搭建以 X-CLIP 等模型为教师、CLIP4Clip 为学生的蒸馏框架，通过帧级、视频级知识蒸馏与注意力帧特征聚合，将教师模型的细粒度跨模态对齐能力迁移至轻量学生模型。
- 主要负责将学生端 ViT 视觉编码器替换为 OpenCLIP 预训练 ConvNeXt，进一步压缩学生模型规模，完成模型接口封装、跨架构权重映射及注意力池化头适配；围绕全量微调、骨干冻结、渐进解冻和池化头优先对齐等策略开展消融实验，研究卷积视觉表征与 CLIP 共享嵌入空间的适配方式。

研究兴趣

- VLM、Multimodal Representation Learning、Video Understanding
- Diffusion Models、Flow Matching
- VLA（Action Chunking / Real-Time Inference）、World Models

专业技能

视觉与多模态： CNN、Transformer、ViT、ConvNeXt、CLIP、Contrastive Learning、Representation Learning
生成模型： DDPM / DDIM、Flow Matching
机器学习与建模： Supervised Learning、Ensembles、Clustering、Dimensionality Reduction、Time-Series Modeling
模型训练与优化： LoRA Fine-tuning、Knowledge Distillation、Distributed Training
框架与工具： PyTorch、Hugging Face、Python、Java、Linux、Slurm、Git、Conda、LaTeX

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Education

Southern University of Science and Technology (SUSTech) Undergraduate, 2024 Cohort
Department of Computer Science and Engineering | Computer Science and Technology Expected Graduation: June 2028
GPA: 3.76 / 4.00 **Weighted Average:** 89.16 / 100

Research Experience

Gaze-Trajectory Generation with Diffusion and Flow Models Aug 2026 – Present
Computational Linguistics and Consciousness Sciences Lab (CLCS), SUSTech

- Researching faster sampling, long-sequence gaze modeling, and cross-dataset adaptation for the DDPM-based ScanDL 2.0.
- Converting DDPM sampling to DDIM and fine-tuning on the OneStop eye-tracking dataset; evaluating trade-offs among sampling steps, generation quality, and inference latency.
- Exploring text-conditioned Flow Matching with ODE integration for efficient few-step gaze-trajectory generation.

Multimodal Video–Text Retrieval and Clip Localization for Driving Scenes Jun 2026 – Aug 2026
Research Institute of Trustworthy Autonomous Systems (RITAS), SUSTech

- Built a bidirectional video–text retrieval system with Qwen3-VL-Embedding, InfoNCE, and LoRA, followed by Qwen3-VL-Reranker Top-K reranking; achieved **89.9%** Text-to-Video and **85.7%** Video-to-Text Recall@1 on 168 validation videos.
- Designed a Transformer trajectory encoder and injected vehicle trajectories into Qwen visual-token hidden states for joint vision–language–trajectory learning; improved single-stage Text-to-Video / Video-to-Text Recall@1 from 80.4%/80.4% to **82.1%/81.5%** (+1.7/+1.1).
- Extended the model to Text-to-Clip localization using label-aware positive/hard-negative construction, similarity filtering, and gap filling; raised mean temporal IoU from 0.333 to **0.406** (+21.9%) and frame-set precision from 42.7% to **48.3%**.

Frequency-Guided Lightweight Small Object Detection Dec 2025 – Mar 2026
Co-first author · Submitted to AAAI · arXiv:2606.23825
Research Institute of Trustworthy Autonomous Systems (RITAS), SUSTech

- Developed a frequency-guided framework using WDG, LGE, and FDHead to preserve small-object boundary information across the backbone, neck, and detection head.
- Adapted the modules to RT-DETRv2's CNN–Transformer pipeline: integrated WDG into YOLO11 P2/P3 C3k2_WDC blocks, added lightweight LGE after the HybridEncoder P4/S4 projection, and designed area-biased Top-K query selection.
- Validated RT-DETR-R18/R50 on VisDrone2019, UAVDT, TinyPerson, and DOTAv1; RT-DETR-R18+ gained **+7.5 mAP**$_{50}$ on UAVDT val while reducing parameters by **31.3%** and GFLOPs by **25.8%**.

Knowledge Distillation for Video–Text Retrieval Sep 2025 – Nov 2025
Research Institute of Trustworthy Autonomous Systems (RITAS), SUSTech

- Built a TeachCLIP distillation framework with X-CLIP as the teacher and CLIP4Clip as the student, transferring cross-modal alignment through frame/video-level distillation and attention-based frame aggregation.
- Replaced the student's ViT with pretrained OpenCLIP ConvNeXt; implemented interface wrappers, cross-architecture weight mapping, and attention-pooling adaptation, and ablated full fine-tuning, backbone freezing, gradual unfreezing, and pooling-head-first alignment.

Research Interests

- VLMs, Multimodal Representation Learning, Video Understanding
- Diffusion Models, Flow Matching
- VLAs (Action Chunking / Real-Time Inference), World Models

Technical Skills

Vision / Multimodal:	CNN, Transformer, ViT, ConvNeXt, CLIP, Contrastive Learning, Representation Learning
Generative Models:	DDPM / DDIM, Flow Matching
Machine Learning:	Supervised Learning, Ensembles, Clustering, Dimensionality Reduction, Time-Series Modeling
Model Training:	LoRA Fine-tuning, Knowledge Distillation, Distributed Training
Frameworks & Tools:	PyTorch, Hugging Face, Python, Java, Linux, Slurm, Git, Conda, LaTeX